An existential definition of the integers in the rationals from reduced arithmetic inputs

Third version: a model-first proof and a direct proof
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Abstract

We give two existential definitions of $\mathbb{Z}$ in $\mathbb{Q}$, conditional on three specified inputs from the OpenAI H10 manuscript: a geometric family, restricted simultaneous point existence, and prime-contact representations. An earlier height theorem of von Känel–Kret replaces the manuscript's arithmetic height engine. Classical descent and formal groups supply the other arithmetic ingredients. A finite certificate first yields a Diophantine model of $\mathbb{Z}$, which already refutes Mazur's finite-component conjecture; DPRM then identifies the actual integers. A separate proof replaces this last use of the model by Lucas growth. All certificates are constructed explicitly as finite existential formulas.

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1 Statement and conventions

Theorem 1.1. *Assume Hypotheses 2.1, 2.4, and 2.5, extracted below from [8]. Then $\mathbb{Z}$ is existentially definable in $\mathbb{Q}$. Moreover, the first stage of the proof constructs a Diophantine model of $\mathbb{Z}$ in $\mathbb{Q}$; this alone contradicts Mazur's conjecture that the closure of the rational points of every variety over $\mathbb{Q}$ has finitely many connected components in its real locus.*

The result uses von Känel–Kret [4], an earlier preprint, in addition to the stated H10 inputs and the published results cited below. We treat those inputs as black boxes; no claim of independent verification of them is made. Neither proof uses the conclusion of H10 or its recursive-theoretic argument.

All variables in existential formulas range over $\mathbb{Q}$; a point variable abbreviates finitely many rational coordinates. Rational constants are allowed and may be cleared to integral coefficients. Nonvanishing is existential, by $a \neq 0 \iff \exists t\ at = 1$. Finite conjunctions and disjunctions of polynomial equations are Diophantine over $\mathbb{Q}$, using sums of squares and products. Thus an existential definition below is equivalent to one polynomial equation with existentially quantified rational variables.

Write $v_p(p) = 1$, $v_p(0) = +\infty$, and $\mathrm{Supp}(a) = \{p : v_p(a) \neq 0\}$ for $a \in \mathbb{Q}^\times$. For places of $F = \mathbb{Q}(\sqrt{2})$ above p use the normalization $v(p) = 1$. Let h denote absolute logarithmic height. For primitive integers u, v, $h((u : v)) = \log \max(|u|, |v|)$. Fix five distinct rational contacts $\mathcal{C} \subset \mathbf{P}^1$ containing ∞, and primitive integral forms $L_b(U, V)$ vanishing at b, with $L_\infty(U, V) = V$. For a finite prime set S and $s = (u : v) \notin \mathcal{C}$, put

$$M_S(s) = \prod_{\substack{q \notin S \text{ prime} \\ v_q(L_b(u,v)) - \min(v_q(u), v_q(v)) \text{ odd for some } b \in \mathcal{C}}} q. \tag{1}$$

These normalized contact depths do not depend on the chosen coordinates.

2 Arithmetic inputs and their reductions

This section states all the information imported from H10. The three hypotheses are sufficient packages, not assertions of logical necessity. In particular, the prime representation hypothesis retains the convenient thirteen-contact formulation of its source. Subsequent lemmas replace the rank, height, and local-witness inputs used in earlier versions.

2.1 The geometric input and the height bound

Hypothesis 2.1 (Geometric family)**.** There are contacts $\mathcal{C}$ as above, a finite prime set S_0 containing 3, a quaternion division algebra $B/\mathbb{Q}$, and constants d_0, C_0 such that, for every $s \in \mathbf{P}^1(\mathbb{Q}) \setminus \mathcal{C}$, there are a number field K_0 and an abelian surface A/K_0 with:

(i) $[K_0 : \mathbb{Q}] \le d_0$, and $K_0/\mathbb{Q}$ is unramified outside $S_0 \cup \mathrm{Supp}(M_{S_0}(s))$;

(ii) A has good reduction at every finite place not over S_0;

(iii) $B \hookrightarrow \mathrm{End}^0_{\overline{\mathbb{Q}}}(A)$, and $A_{\overline{\mathbb{Q}}}$ is isogenous to each of its $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-conjugates;

(iv) $h(s) \le C_0(1 + \max(0, h_F(A)))$, where h_F is stable Faltings height.

This is the early geometric part of [8, §4]: its fixed quaternionic family, fiber-field lemma, common-base isogenies, and base-height comparison (source labels `hei:fiber-field` and `hei:base-height`). In particular, (iii) only assumes a *geometric* quaternion action. The homomorphism-field lemma, endomorphism dichotomy, and later arithmetic height argument are not part of this input.

Proposition 2.2. *Hypothesis 2.1 implies, for a fixed $H \ge 1$,*

$$h(s) \le H M_{S_0}(s)^{24} \qquad (s \notin \mathcal{C}). \tag{2}$$

Proof. Put $K = K_0(A[3])$. Silverberg's field-of-definition theorem, in the form [5, Lemma 7.2(b)], makes every geometric endomorphism rational over K, and $[K : K_0] < 3^{16}$. By the Néron–Ogg–Shafarevich criterion [9, Theorem 1], this extension introduces no ramification outside S_0. For $U = \mathrm{Spec}\, \mathcal{O}_K[S_0^{-1}]$, let N_U be the product of the norms of its omitted prime ideals and D_K the absolute discriminant. Good reduction extends A to an abelian scheme over U, and

$$\mathrm{rad}(D_K N_U) \mid \left(\prod_{p \in S_0} p \right) M_{S_0}(s). \tag{3}$$

Suppose first that $A_{\overline{\mathbb{Q}}}$ is not of CM type. Choose a quadratic field $F_0 \subset B$, giving A its GL_2-type structure. Wu [11, Propositions 1.5(1) and 1.7] implies $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A) = B$: the nonsimple non-CM alternative would have endomorphism algebra $M_2(\mathbb{Q})$, which cannot contain the division algebra B; in the simple surface case the noncommutative alternative is a quaternion algebra over $\mathbb{Q}$, hence equals B. Each conjugacy isogeny transports B by an algebra automorphism; Skolem–Noether corrects this by a quasi-endomorphism to make the isogeny B-compatible. Clearing denominators gives an actual isogeny. Thus A/U is of GL_2-type with $G_{\mathbb{Q}}$-isogenies in the precise sense of [4, (7.1)]. Equation (7.19) there gives

$$h_F(A) \le 4(6\ell)^{288\ell} \mathrm{rad}(D_K N_U)^{24}, \qquad \ell = ([K : \mathbb{Q}]3^{16})!.$$

The prefactor is uniformly bounded. If $A_{\overline{\mathbb{Q}}}$ is of CM type, [4, Theorem 7.1(ii)] instead gives $h_F(A) \le 6^{100} \mathrm{rad}(D_K N_U)^{10}$. Combine these estimates with (3) and (iv). □

This deduction replaces H10's cocycle splitting, corrected Galois representations, Fontaine–Mazur input, conductor estimates, and modular-Jacobian comparison. The earlier theorem [4] is a preprint result; the published rational-base height theorem does not by itself supply the varying-number-field statement used here.

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2.2 The elliptic curve

Fix

$$F = \mathbb{Q}(\sqrt{2}), \qquad c = 3 - \sqrt{2}, \qquad E:\ y^2 = x^3 - c^2 x, \qquad z = -x/y. \tag{4}$$

Write σ for the nontrivial automorphism of $F/\mathbb{Q}$.

Lemma 2.3. *For the curve (4), $E(F)$ has rank one, and $Q_0 = (1+\sqrt{2}, 2\sqrt{2})$ is nontorsion.*

Proof. In this proof, finite-place valuations have value group $\mathbb{Z}$. The Minkowski bound $\sqrt{8}/2 < 2$ shows that $\mathcal{O}_F = \mathbb{Z}[\sqrt{2}]$ is principal. Its units modulo squares have order four, and $-1, 1+\sqrt{2}$ realize all real sign patterns; hence every totally positive unit is a square. Also c is totally positive and has norm 7, so (c) is a prime with residue field $\mathbf{F}_7$.

For the two-isogenous curve $E' : y^2 = x^3 + 4c^2x$, let $\alpha : E(F) \to F^\times/F^{\times 2}$ and $\alpha' : E'(F) \to F^\times/F^{\times 2}$ be the usual descent maps, given by the x-coordinate away from $O, (0,0)$. Their exceptional values are $1, -c^2$ and $1, 4c^2$, respectively [10, Example X.4.8 and Proposition X.4.9]. A nonzero x on E has even valuation away from (c): if its valuation is nonzero, precisely one of $x^3, -c^2x$ has smaller valuation, which would be odd if $v(x)$ were odd. Consequently $|\operatorname{im}\alpha| \le 8$. The two-torsion points give the distinct classes $1, -1, c, -c$. Since

$$(1+\sqrt{2})^3 - c^2(1+\sqrt{2}) = 8,$$

the point Q_0 supplies a class with opposite real signs, so $|\operatorname{im}\alpha| = 8$.

Every nonzero x on E' is totally positive. Its finite valuations are even away from $2, (c)$ by the same argument. At the prime above 2, in the integer-valued normalization, $v(4c^2) = 4$; for odd $v(x)$ the numbers $3v(x)$ and $4 + v(x)$ are unequal and have odd minimum. At (c) the same comparison leaves only $v(x) = 1$; writing $x = cu$ would then require $u^2 \equiv -4 \pmod{(c)}$, impossible in $\mathbf{F}_7$. Thus all finite valuations are even, and principality and total positivity make x a square. Hence $\operatorname{im}\alpha' = \{1\}$. The two-isogeny exact sequence [10, Remark X.4.7] now gives

$$2^{\operatorname{rank} E(F)+2} = |\operatorname{im}\alpha|\,|\operatorname{im}\alpha'| = 8;$$

here the two-isogeny maps $E[2]$ onto the kernel of its dual. For the finite torsion subgroup T, one has $|\alpha(T)| \le |T/2T| = |T[2]| = 4$. Its image is therefore exactly the four classes already supplied by $E[2]$, and Q_0 is nontorsion. □

This curve is isomorphic to the original H10 curve: with $u = 3 - 2\sqrt{2}$ one has $75 - 53\sqrt{2} = cu^2$, and the isomorphism is $(x, y) \mapsto (u^2x, u^3y)$. Since u is a global unit, it preserves all reduction orders and formal-parameter valuations. By Mordell–Weil [10, Theorem VIII.6.7], choose an even positive integer m_0 annihilating the torsion and a generator P_0 of the infinite cyclic group $m_0E(F)$. In particular $P_0 \in 2E(F)$.

2.3 Prime-contact representations

Let $\mathscr{P}$ be the split rational primes $q > 25$ of good reduction for E, E^σ for which, at either place over q,

$$d_1 = \#\widetilde{E}(\mathbb{F}_q), \qquad d_2 = \#\widetilde{E^\sigma}(\mathbb{F}_q), \qquad \gcd(d_1, d_2) = 4.$$

Changing the chosen place interchanges d_1, d_2.

Hypothesis 2.4 (Prime representations)**.** There is a finite set $\mathcal{K} \subset \mathbb{Z}_{>0}$ such that, for every $A \in \mathbb{Z}$ and every prime lower bound, there are integers u_1, u_2, u_3, v, with $v \neq 0$ and $u_1 + u_2 + u_3 = Av$, for which each of

$$v, \qquad L_b(u_i, v) \quad (1 \le i \le 3,\ b \in \mathcal{C} \setminus \{\infty\}) \tag{5}$$

is $\varepsilon\kappa q$, with $\varepsilon \in \{\pm 1\}$, $\kappa \in \mathcal{K}$, and $q \in \mathscr{P}$ exceeding that bound.

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This is [8, Lemma 6.1], transported across the unit isomorphism above. It is independent of the chosen multiple of P_0. Its proof uses prime patterns and an additional sieve imposing the group-order condition; the combined conclusion is retained as a black box. Formal-group witnesses are not part of this hypothesis: they will follow by the Chinese remainder theorem in Lemma B.2.

2.4 The point-existence input

For $a > 0$ rational, let $\mathrm{sf}(a)$ be the positive squarefree integer representing $a\mathbb{Q}^{\times 2}$, and put

$$E_l : \quad y^2 = x(x-l)(x+3l) \qquad (l \in \mathbb{Q}^\times).$$

Fix finite sets $\mathcal{M} \subset \mathbb{Q}_{>0}$ and $\mathcal{D}_m \subset \mathbb{Q}_{>0}$, for $m \in \mathcal{M}$, with these properties: for every $b > 0$, some $m \in \mathcal{M}$ makes mb a square in $\mathbb{Q}_2$ and a unit nonsquare in $\mathbb{Q}_3$; for every m and $b > 0$, some $m_d \in \mathcal{D}_m$ makes $m_d b$ a square in every $\mathbb{Q}_p$ with $p \in S(m) := \{2,3\} \cup \mathrm{Supp}(m)$. Here "unit nonsquare" specifies a squareclass, so its valuation is allowed to be any even integer. These finite lists exist by weak approximation and the finiteness and openness of the local squareclasses [7, Theorems 7.20 and 7.32]. Indeed, $1 + p\mathbb{Z}_p$ consists of squares for odd p, as does $1 + 8\mathbb{Z}_2$; valuation parity and finitely many unit residues therefore determine the squareclasses. Include the real place when selecting positive rational representatives. Let S_1 contain $2, 3$ and the supports of all members of these lists.

Hypothesis 2.5 (Restricted simultaneous point existence)**.** Let $q \in \mathscr{P} \setminus S_1$, $m \in \mathcal{M}$, and $\beta = \mathrm{sf}(mq)$. Suppose β is a square in $\mathbb{Q}_2$ and a unit nonsquare in $\mathbb{Q}_3$. For every positive squarefree integer θ satisfying

$$\theta \in \mathbb{Q}_p^{\times 2} \quad (p = 2, 3 \text{ or } p \mid \beta), \qquad \beta \in \mathbb{Q}_p^{\times 2} \quad (p \mid \theta),$$

there is $e \in \mathbb{Q}^\times$ such that both E_e and $E_{e\theta}$ have a rational point with nonzero x-coordinate in $\beta\mathbb{Q}^{\times 2}$.

This is the restricted case of the simultaneous matrix and point realization assertions in [8, Lemmas 3.2, 3.4, 3.6]. Those assertions produce a positive squarefree e; no such restriction on e is used here. The displayed conditions imply that the supports of β, θ are disjoint and avoid $2, 3$, and that $\beta \neq 1$. The first squareclass has only one variable prime.

2.5 Fixed constants and classical local facts

Put $K = 24$. Choose a finite prime set S containing S_0, S_1, all primes at most $K + 1$, all ramified and bad-reduction primes, the supports of $\mathcal{K}$, and the prime divisors of the leading coefficients of the finite contact forms. With

$$H_d = \max\left\{1, H\left(\prod_{q \in S \setminus S_0} q\right)^{24}\right\}, \qquad h(s) \le H_d M_S(s)^{24} \quad (s \notin \mathcal{C}).$$

The canonical height satisfies $|h(x(R)) - 2\widehat{h}(R)| \le C_E$ for all $R \neq O$ on this fixed curve, with $C_E \ge 0$, and $\widehat{h}(nR) = n^2\widehat{h}(R)$ [10, Theorem VIII.9.3(b),(d),(e)]. Choose a positive integer N so large that, with

$$P = NP_0, \quad m = m_0 N, \quad G = mE(F) = \mathbb{Z}P, \qquad C = \frac{C_E + 9H_d + 3\log 3 + \frac{3}{2}\log 2}{2\widehat{h}(P)}, \qquad 0 < C < \tfrac{1}{4}. \tag{6}$$

Such a choice is possible by quadraticity and positivity of canonical height. No exceptional primes are added when P_0 is rescaled; in particular H_d is independent of N. Notice that $P \in 2E(F)$. All constants from now on are fixed, independently of every quantified variable.

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We use unique point codes

$$\mathscr{E} = \{(0,0,0,0,0)\} \cup \{(1, x_0, x_1, y_0, y_1) : (y_0 + y_1\sqrt{2})^2 = (x_0 + x_1\sqrt{2})^3 - c^2(x_0 + x_1\sqrt{2})\}.$$

The first code is O. Equalities in F mean equality of both rational coefficients. Algebraicity of the group law [10, Theorem III.3.6], on finitely many affine charts, gives existential graphs for addition and each fixed multiplication map. Thus $G(Q) \iff \exists R \in \mathscr{E}\ [m]R = Q$ is existential. On G put

$$f(Q) = 1/x(Q), \quad z(Q) = -x(Q)/y(Q) \quad (Q \neq O), \qquad f(O) = z(O) = 0.$$

Every nonzero point of G is nontorsion, so $xy \neq 0$. All rational expressions below mean their existential graphs, with denominators nonzero: $Y = A/B$ in F abbreviates $BY = A$ and $t\,\mathrm{N}(B) = 1$ for some $t \in \mathbb{Q}$. This also applies to intermediate denominators in nested expressions.

We shall use two standard local consequences. First, a nonzero point is formal at a good place exactly when $v(f(Q)) > 0$. Second, given finitely many finite places and prescribed depths, a positive multiple LP lies at those depths in every formal group, as does every nonzero integer multiple of LP. Indeed the formal subgroups form a basis of open subgroups of $E(F_v)$, compact by projectivity, and hence have finite index [10, Proposition VII.2.2 and Theorem IV.6.4]; take L divisible by those indices. The functions f, z vanish at O, of orders two and one, respectively. Finally, for odd q, if $D = A + B\sqrt{2}$ has valuation at least T at every place over q, then $v_q(A), v_q(B) \geq T$: use $A = (D + D^\sigma)/2$ and $B = (D - D^\sigma)/(2\sqrt{2})$.

3 Two common existential certificates

3.1 A restricted valuation test

Appendix A constructs an existential predicate Φ with the following two properties:

$$\begin{aligned} &\Phi(b),\ p \notin S_1,\ v_p(b) < 0 \quad &&\Longrightarrow v_p(b) \in 2\mathbb{Z}, \\ &q \in \mathscr{P} \setminus S_1,\ v_q(z) \geq 0 \quad &&\Longrightarrow \Phi(qz^2). \end{aligned} \tag{7}$$

Also $\Phi(0)$ holds. Only the second implication uses Hypothesis 2.5; the first is elementary. In particular no completeness assertion on all square-denominator rationals is needed. For a fixed positive integer L, define

$$\mathsf{G}_L(r, w): \quad r \neq 0 \ \wedge\ \Phi(r) \ \wedge\ \Phi(w^2/r^{2L-1}).$$

Lemma 3.1. *If $\mathsf{G}_L(r, w)$ holds, $q \notin S$, and $v_q(r)$ is odd, then $v_q(r) > 0$ and $v_q(w) \geq L$. Conversely, for $q \in \mathscr{P} \setminus S$ and $v_q(w) \geq L$, one has $\mathsf{G}_L(q, w)$.*

Proof. The first test excludes negative odd $v_q(r)$. If $w \neq 0$, the odd valuation $2v_q(w) - (2L-1)v_q(r)$ cannot be negative, forcing $v_q(w) \geq L$; $w = 0$ is immediate. Conversely, apply (7) to q and $q(w/q^L)^2$. □

3.2 Comparing rational proxies with elliptic indices

Proposition 3.2. *For every fixed positive integer k there is a positive existential predicate $\mathsf{T}_K^{(k)}(\mathbf{Q}; \mathbf{a})$ on $(G \setminus \{O\})^k \times \mathbb{Q}^k$ with these properties. Write $Q_i = n_i P$ and $B = \max_i |n_i|$.*

(i) *It holds whenever $a_i = n_i$ for all i.*

(ii) *If it holds and $F \in \mathbb{Z}[T_1, \ldots, T_k]$ satisfies $F(\mathbf{a}) = 0 \neq F(\mathbf{n})$, then $B^2 \leq C|F(\mathbf{n})|$.*

(iii) *If it holds and $a_i \neq n_i$, then $B^2 \leq C|\,\mathrm{num}(a_i - n_i)|$, where* num *is the reduced numerator.*

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The full construction is in Appendix B. Each rational coordinate of the points is represented by three slopes. At primes with odd contact depth, finite local tests force every proxy to agree with its index modulo q^{24}. A nonzero error is therefore divisible by the 24th power of every relevant contact radical. The height bound and quadratic canonical height give (ii) and (iii). The constant C is independent of k; only arities 3 and 6 are used.

3.3 A scalar height certificate

For $\beta \in \mathbb{Q}$ and $Q_s \in G \setminus \{O\}$ define $\widetilde{\mathsf{H}}_K(\beta, Q_s)$ as follows. Quantify rational u_1, u_2, u_3, v_0, with $v_0 \neq 0$ and $u_1 + u_2 + u_3 = \beta v_0$. For each of its thirteen contacts (5), require

$$c_\gamma = \varepsilon \kappa r_\gamma \neq 0, \qquad \varepsilon \in \{\pm 1\}, \quad \kappa \in \mathcal{K},$$

as a finite disjunction, and choose a fresh $J_\gamma \in G \setminus \{O\}$. Apply $\mathsf{G}_1(r_\gamma, \cdot)$ to both rational coefficients of $f(Q_s)$ and of $f(J_\gamma)$, and apply $\mathsf{G}_K(r_\gamma, \cdot)$ to both coefficients of $z(Q_s)/z(J_\gamma)$. All witnesses are rational point codes or rational numbers; this defines a finite existential formula.

Lemma 3.3. *Writing $Q_s = sP$, this certificate implies*

$$h(\beta) \leq 3H_d|s| + \log 3. \tag{8}$$

For every integer β it holds for arbitrarily large positive s.

Proof. At any $q \notin S$ of odd raw-contact valuation, r_γ also has odd valuation, because κ is a q-unit. The coefficient tests force Q_s, J_γ to be formal at every place over q, and their parameter ratio to have valuation at least K. Formal comparison (Lemma B.1) gives $v_q(s/j_\gamma) \geq K$, where $J_\gamma = j_\gamma P$; since j_γ is an integer, $q^K \mid s$. Every prime in a normalized contact radical occurs in some raw contact (Lemma B.5). Each slope therefore has height at most $H_d|s|^{24/K} = H_d|s|$. The height inequality for the sum of three numbers proves (8).

For integer β, Hypothesis 2.4 supplies the contacts with $r_\gamma = q \in \mathscr{P} \setminus S$. Choose each J_γ sufficiently deep at both places over its prime, and a common positive multiple $Q_s = sP$ sufficiently deep at all these places that its ratios have depth at least K. The local facts in Section 2 give such multiples with unbounded positive indices. They also give the asserted depths for both rational coefficients. Lemma 3.1 supplies all required tests. □

4 First proof: a model, Mazur's conjecture, and the actual integers

Proposition 4.1. *The map $n \mapsto nP$ is a Diophantine model of the ring $\mathbb{Z}$ in $\mathbb{Q}$.*

Proof. The domain G and its addition are existential. For nonzero $U, V, W \in G$, put

$$\mathsf{M}(U, V, W) \iff \exists a, b, c \in \mathbb{Q}\, [\mathsf{T}_K^{(3)}((U, V, W); (a, b, c)) \wedge ab = c].$$

Write $(U, V, W) = (nP, mP, kP)$ and $B = \max(|n|, |m|, |k|)$. True multiplication satisfies this formula by completeness. If it holds but $nm \neq k$, Proposition 3.2(ii) gives

$$B^2 \leq C|nm - k| \leq 2CB^2 < B^2,$$

a contradiction. Add the branch $(U = O \vee V = O) \wedge W = O$, and explicitly restrict all three points to G. This defines multiplication everywhere; the ring constants are O, P. □

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Corollary 4.2 (Model-only consequence)**.** *Under the stated arithmetic inputs, Mazur's finite-component conjecture is false.*

Proof. Cornelissen–Zahidi [1, Theorem 3.6] proves that this conjecture excludes a Diophantine model of $\mathbb{Z}$ in $\mathbb{Q}$. Their definition permits a domain in $\mathbb{Q}^k$, so it applies to our point codes. □

Thus the contradiction to Mazur requires only Proposition 4.1. Neither the scalar certificate nor either subsequent definition of the actual copy of $\mathbb{Z}$ is needed for it.

To pass from the model to the actual integers, choose a positive integer L with $L \log 2 > 3H_d + 1$. By DPRM [2, 6], the recursive relation

$$\Xi(sP, tP) \quad \Longleftrightarrow \quad s \geq 1, \quad t = 2^{Ls}$$

is existential on the model. Explicitly, translate a Diophantine definition over $\mathbb{Z}$ by replacing every variable by a point constrained to G, every operation by its model graph, and every integer constant by its fixed point. This uses only finitely many existential variables.

For $X \in \mathbb{Q}$ put $\beta = X^2 + 2$ and require

$$\begin{aligned} \exists A, Q_s, Q_t \in G \setminus \{O\}\ \exists \xi, \tau \in \mathbb{Q} : \quad & \mathsf{T}_K^{(3)}((A, Q_s, Q_t); (\beta, \xi, \tau)) \\ & \wedge\, \widetilde{\mathsf{H}}_K(\beta, Q_s) \wedge \Xi(Q_s, Q_t). \end{aligned} \tag{9}$$

If this holds, write $(A, Q_s, Q_t) = (nP, sP, tP)$ and $B = \max(|n|, s, t)$. If $\beta \neq n$, its reduced numerator satisfies $|\operatorname{num}(\beta - n)| \leq 2B \exp h(\beta)$. Proposition 3.2(iii) and Lemma 3.3 give

$$\log B \leq \log(2C) + h(\beta) < 3H_d s + \log(3/2).$$

But $B \geq t = 2^{Ls}$ gives $\log B > (3H_d + 1)s$, impossible for $s \geq 1$. Thus $\beta = n \in \mathbb{Z}$. A rational number whose square is an integer is an integer, so $X \in \mathbb{Z}$. Conversely, for $X \in \mathbb{Z}$ take $n = X^2 + 2$, any positive scalar-certificate index s, and $t = 2^{Ls}$; true indices as proxies satisfy (9). This proves the first definition.

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5 Second proof: a scalar height bound and Lucas growth

This proof uses the finite certificate directly, without constructing integer multiplication or invoking DPRM. Throughout, $K = 24$, $C < 1/4$, and $H_d \geq 1$ have the fixed values chosen above. Recall that $\widetilde{\mathsf{H}}_K(\beta, sP)$ implies

$$h(\beta) \leq 3H_d|s| + \log 3,$$

and that, for every integer β, this predicate holds for arbitrarily large positive integers s.

Lemma 5.1 (Lucas growth)**.** *Let $s, t, w \in \mathbb{Z}$, with $a = |s| \geq 7$ and $w \neq 0$. If*

$$t^2 - s(s-2)tw + (s-2)^2 w^2 = 1, \tag{10}$$

then $|t| \geq (a-1)^{a-4}$. For every integer s with $|s| \geq 7$, there are positive integers t, w satisfying (10).

Proof. Put $y = \operatorname{sgn}(s)(s-2)w$. Equation (10) becomes $t^2 - aty + y^2 = 1$, or

$$(2t - ay)^2 - (a^2 - 4)y^2 = 4.$$

Define $U_0 = 0$, $U_1 = 1$, and $U_{j+1} = aU_j - U_{j-1}$. Jones's classification of nonnegative solutions of this Pell equation [3, Corollary 2.7], applied to $(|2t - ay|, |y|)$, gives an integer $k \geq 1$ with $|y| = U_k$ and $|2t - ay| = V_k$. Here V_k is the companion Lucas sequence. The identities

$$2U_{k+1} = aU_k + V_k, \qquad 2U_{k-1} = aU_k - V_k$$

[3, (5)–(6)] therefore give $|t| \in \{U_{k-1}, U_{k+1}\}$, for either sign of y or $2t - ay$. Induction in the recurrence gives

$$U_j \equiv j \pmod{a-2}, \qquad U_j \equiv (-1)^{j-1} j \pmod{a+2}, \qquad (j \geq 1),$$
$$U_{j+1} \geq (a-1)U_j, \qquad U_j \geq (a-1)^{j-1} \qquad (j \geq 1).$$

If $s > 0$, then $a - 2$ divides $U_k = |y|$, so $a - 2$ divides k. If $s < 0$, then $a + 2$ divides U_k, so $a + 2$ divides k. In either case $k \geq a - 2$, whence

$$|t| \geq U_{k-1} \geq (a-1)^{k-2} \geq (a-1)^{a-4}.$$

For existence, take $k = a - 2$ when $s > 0$ and $k = a + 2$ when $s < 0$, and set $t = U_{k+1}$ and $w = U_k/k$. The congruences make w a positive integer; in both cases $y = U_k$. The recurrence identity $U_{k+1}^2 - aU_{k+1}U_k + U_k^2 = 1$ verifies (10). □

Fix an integer

$$N_* > \max\{8, 1 + \exp(8H_d)\}.$$

For $X \in \mathbb{Q}$, put $\beta = X^2 + 2$. Define $\mathsf{Z}_{\mathrm{dir}}(X)$ by the existence of six points

$$(Q_n, Q_s, Q_t, Q_w, Q_v, Q_p) \in (G \setminus \{O\})^6$$

and rational numbers $\xi, \tau, \omega, \nu, \pi$ satisfying

$$\mathsf{T}_K^{(6)}((Q_n, Q_s, Q_t, Q_w, Q_v, Q_p); (\beta, \xi, \tau, \omega, \nu, \pi)),$$
$$\widetilde{\mathsf{H}}_K(\beta, Q_s), \qquad \bigwedge_{j=-N_*}^{N_*} Q_s \neq jP, \tag{11}$$
$$\nu = (\xi - 2)\omega, \qquad \pi = \xi\nu, \qquad \tau^2 - \pi\tau + \nu^2 = 1.$$

All the conditions are positive existential over $\mathbb{Q}$: $G = \mathbb{Z}P$ is already existentially defined, and the point inequalities form a fixed finite conjunction.

Theorem 5.2. *For every $X \in \mathbb{Q}$, one has $\mathsf{Z}_{\mathrm{dir}}(X)$ if and only if $X \in \mathbb{Z}$.*

Proof. Suppose the formula holds. Write the six points as (nP, sP, tP, wP, vP, pP), and put

$$B = \max\{|n|, |s|, |t|, |w|, |v|, |p|\}, \qquad a = |s| > N_*.$$

The residuals of the three quadratic equations in (11), evaluated at the integer indices, have absolute values at most

$$B^2 + 3B, \qquad B^2 + B, \qquad 3B^2 + 1,$$

respectively, hence at most $4B^2$. If any residual were nonzero, the polynomial-error clause of Proposition 3.2 would give $B^2 \leq 4CB^2 < B^2$. Consequently

$$v = (s-2)w, \qquad p = sv, \qquad t^2 - pt + v^2 = 1,$$

so (10) holds. Since $w \neq 0$, Lemma 5.1 gives

$$\log B \geq (a-4)\log(a-1) > 4H_d a. \tag{12}$$

The strict inequality uses $a > 8$ and $\log(a-1) > 8H_d$.

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If $\beta \neq n$, let $D \neq 0$ be the reduced numerator of $\beta - n$. Then $|D| \leq 2B\exp(h(\beta))$, so the numerator-error clause of Proposition 3.2 gives

$$B^2 \leq C|D| \leq 2CB\exp(h(\beta)), \qquad \log B \leq \log(2C) + h(\beta) < 3H_d a + \log(3/2).$$

This contradicts (12), since $H_d \geq 1$ and $a > 8$. Thus $\beta = n \in \mathbb{Z}$. Now $X^2 = \beta - 2 \in \mathbb{Z}$, and a rational number whose square is an integer is itself an integer.

Conversely, suppose $X \in \mathbb{Z}$. Then $\beta = X^2 + 2 > 0$ is integral. Choose a positive $s > N_*$ with $\widetilde{\mathsf{H}}_K(\beta, sP)$, and choose positive t, w from Lemma 5.1. Set

$$n = \beta, \qquad v = (s-2)w, \qquad p = sv.$$

All six indices are nonzero. Taking the proxies to be the actual indices makes the three quadratic equations true, and completeness of $\mathsf{T}_K^{(6)}$ supplies its witnesses. Hence $\mathsf{Z}_{\mathrm{dir}}(X)$ holds. □

A Construction of the parity predicate

Define $\Phi(b)$ to hold if $b = 0$, or if $b \neq 0$ and, for some $m \in \mathcal{M}$, $m_d \in \mathcal{D}_m$, and $t, e \in \mathbb{Q}$, put

$$b' = mb, \qquad d_b = m_d b, \qquad U = b\frac{1 - d_b t^2}{1 + d_b t^2}, \qquad H' = b'(b' - U^2), \tag{13}$$

and require $1 + d_b t^2 \neq 0$, $eH' \neq 0$, and a rational point with nonzero x-coordinate in $b'\mathbb{Q}^{\times 2}$ on each of $E_e, E_{eH'}$. This is an existential ring formula: the lists give finite disjunctions, the point conditions use $x = b'u^2$ with $u \neq 0$, and nonvanishing and rational-function graphs are encoded in Section 2. The local normalizations used to choose the lists are not additional conjuncts of Φ.

Proposition A.1. *The predicate Φ has the following properties.*

(i) *If $\Phi(b)$ holds and $p \notin S_1$, then $v_p(b) < 0$ implies $v_p(b) \in 2\mathbb{Z}$.*

(ii) *Assuming Hypothesis 2.5, one has $\Phi(qz^2)$ whenever $q \in \mathscr{P} \setminus S_1$ and $z \in \mathbb{Q}$ satisfies $v_q(z) \geq 0$.*

Proof. For (i), suppose $k = v_p(b) < 0$ is odd. All multipliers are p-units. If $t \neq 0$, the odd valuation of $d_b t^2$ is nonzero, so $(1 - d_b t^2)/(1 + d_b t^2)$ is a unit; for $t = 0$ it is 1. Hence

$$v_p(U) = k, \qquad v_p(H') = 3k.$$

One of e, eH' has even valuation, say $l = p^{2a}L$ with L a p-unit. Scaling its point gives $Y^2 = X(X - L)(X + 3L)$ and preserves the x-squareclass. If $v_p(X) < 0$, the right side has valuation $3v_p(X)$; if $v_p(X) > 0$, it has valuation $v_p(X)$, since $p \neq 3$. Thus every nonzero X has even valuation, contradicting its squareclass $b'\mathbb{Q}_p^{\times 2}$.

For (ii), the case $z = 0$ is immediate. Put $b = qz^2 > 0$ and choose m, m_d with the stated normalization properties. At the real place and at every $p \in S(m)$, approximate a solution of $d_b t^2 = 1$. Weak approximation and openness of nonzero local squares give a rational t for which $1 + d_b t^2 \neq 0$ and

$$H' > 0, \qquad H' = b'^2(1 - U^2/b') \in \mathbb{Q}_p^{\times 2} \quad (p \in S(m)).$$

Write $\beta = \mathrm{sf}(b') = \mathrm{sf}(mq)$, $\theta = \mathrm{sf}(H')$, and $B = \mathrm{Supp}(\beta)$, $J = \mathrm{Supp}(\theta)$. Then $B \subseteq S(m) \cup \{q\}$. At q, the integer $v_q(b) = 1 + 2v_q(z)$ is positive and odd; since $q \notin S_1$, the same unit-quotient calculation gives

$$v_q(U) = v_q(b), \qquad v_q(U^2/b') = v_q(b) > 0.$$

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Thus H' is a square in $\mathbb{Q}_q$, and hence at every prime in B. It follows that $B \cap J = \varnothing$.

If $U = 0$, then $H' = b'^2$ and $J = \varnothing$. Otherwise, for $p \in J$, both $v_p(b')$ and $2v_p(U)$ are even, while $v_p(H')$ is odd. Consequently they are equal, with strict cancellation in $b' - U^2$. Since $p \notin S(m)$, it is odd, and

$$b'/U^2 \in 1 + p\mathbb{Z}_p \subseteq \mathbb{Q}_p^{\times 2}.$$

Therefore β is a square in $\mathbb{Q}_p$ for every $p \in J$. The required conditions at $2, 3$ hold by construction. Hypothesis 2.5 supplies the two points. Writing $H' = \theta c^2$ with $c \in \mathbb{Q}^\times$, the map $(x, y) \mapsto (c^2 x, c^3 y)$ sends $E_{e\theta}$ to $E_{eH'}$ and preserves the x-squareclass. These are witnesses for $\Phi(b)$. □

B Construction of the finite certificate

We use the fixed data of Proposition 3.2 and the point codes of Section 2. All rational-function expressions below use their existential graphs, with nonzero denominators.

Lemma B.1 (Uniform formal comparison)**.** *Let $q \notin S$ and let $v = v_\mathfrak{q}$ at a place above q, normalized by $v(q) = 1$. Suppose $U = uP, V = v_0 P$ reduce to O on the displayed integral model, where $u, v_0 \in \mathbb{Z}$ and $v_0 \neq 0$. Then*

$$v(z(U)/z(V)) = v_q(u/v_0). \tag{14}$$

If $v_q(u/v_0) \geq 0$, the truncation ℓ_K of the formal logarithm $\ell(Z) = Z + \sum_{j \geq 2} c_j Z^j \in F[[Z]]$ through degree K satisfies

$$\ell_K(z(V)) \neq 0, \qquad v\left(\frac{\ell_K(z(U))}{\ell_K(z(V))} - \frac{u}{v_0}\right) \geq K. \tag{15}$$

Both conclusions hold for the conjugate curve and point. The case $u = 0$ is included, with $z(O) = \ell_K(0) = 0$.

Proof. The formal logarithm and its valuation-preserving homomorphism property are classical [10, Proposition IV.5.5, Lemma IV.6.3, and Theorem IV.6.4]. At our odd unramified good places, its coefficient of Z^j has valuation at least $-v_q(j)$, and formal parameters have integral valuation at least one. Its homomorphism property gives $v_0 \ell(z(U)) = u\ell(z(V))$, proving (14).

For $j > K$ and $q > K + 1$, one has $j - 1 - v_q(j) \geq K$. Consequently $E_K(Z) = \ell(Z) - \ell_K(Z)$ satisfies $v(E_K(Z)) \geq v(Z) + K$ when $v(Z) \geq 1$, while $v(\ell_K(Z)) = v(Z)$. Put $a = u/v_0$. The exact identity

$$\ell_K(z(U)) - a\ell_K(z(V)) = aE_K(z(V)) - E_K(z(U))$$

has right side of valuation at least $v(z(V)) + K$: the first term has this bound because $v_q(a) \geq 0$, and the second because (14) gives $v(z(U)) \geq v(z(V))$. Division by $\ell_K(z(V))$, of valuation exactly $v(z(V))$, proves (15). If $U = O$, the assertions follow directly. □

Lemma B.2 (Local witnesses from the Chinese remainder theorem)**.** *Suppose $P \in 2E(F)$ has infinite order, and let $q \notin S$ be a prime supplied by Hypothesis 2.4. At either chosen place $\mathfrak{q} \mid q$, there are nonzero integers h, j, with $h \equiv 1 \pmod{q^K}$, such that hP, jP^σ, and $(h-1)P^\sigma$ lie in their formal kernels and*

$$v_\mathfrak{q}\left(\frac{z((h-1)P^\sigma)}{z(jP^\sigma)}\right) \geq K.$$

The same h, j give, for every $a \in \mathbb{Z}$,

$$v_\mathfrak{q}\left(\frac{z((h-1)aP^\sigma)}{z(jP^\sigma)}\right) \geq K, \qquad v_\mathfrak{q}\left(\frac{\ell_K(z(haP))}{\ell_K(z(hP))} - a\right) \geq K,$$

and the two displayed denominators are nonzero. Here $z(O) = \ell_K(0) = 0$ and $v_\mathfrak{q}(0) = +\infty$.

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Proof. Let e_1, e_2 be the orders of the reductions of P, P^σ. Both reduced curves have all four 2-torsion points rational over $\mathbb{F}_q$. Thus the image of doubling in either reduction group has cardinality $d_i/4$. Since P and P^σ are doubles,

$$e_i \mid d_i/4, \qquad \gcd(e_1, e_2) = 1.$$

The Hasse bound [10, Theorem V.1.1] and $q > K + 1$ imply $d_i < 4q$. Because $4 \mid d_i$ and q is odd, this gives $q \nmid d_i$, and hence $q \nmid e_1 e_2$. The Chinese remainder theorem therefore provides a positive h satisfying

$$h \equiv 0 \pmod{e_1}, \qquad h \equiv 1 \pmod{e_2 q^K}.$$

Take $j = e_2$. These congruences give all three formal-kernel conditions and

$$v_q((h-1)/j) \geq K.$$

The formal logarithm preserves valuation on these formal kernels and is a homomorphism. Consequently its valuation identity for index ratios gives the first two asserted inequalities, including $a = 0$ or $h - 1 = 0$ under the stated conventions. Alternatively, adding a sufficiently large multiple of $e_1 e_2 q^K$ makes $h > 1$. Lemma B.1, since $q > K + 1$, gives the last inequality. Its denominator is nonzero because $v_{\mathfrak{q}}(\ell_K(z(hP))) = v_{\mathfrak{q}}(z(hP)) < \infty$; the other denominator is nonzero because jP^σ is nontorsion. The construction is independent of a. □

B.1 Finite local predicates

For $D = A + B\sqrt{2} \in F$ put

$$\mathsf{L}_L(r, \theta; D): \quad \mathsf{G}_L(r, (\theta^2 - 2)B) \wedge \mathsf{G}_L(r, A + B\theta).$$

Lemma B.3 (One common place)**.** *Fix a finite list $D_i = A_i + B_i\sqrt{2} \in F$ and positive integers L_i. If*

$$\mathsf{G}_1(r, \theta^2 - 2) \wedge \bigwedge_i \mathsf{L}_{L_i}(r, \theta; D_i)$$

holds, then for every $q \notin S$ with odd $v_q(r)$ there is one place $\mathfrak{q} \mid q$ such that $v_{\mathfrak{q}}(D_i) \geq L_i$ for every i. Conversely, if $q \in \mathscr{P} \setminus S$ and these inequalities hold at one place above q, then the displayed condition holds with $r = q$ and a suitable rational θ.

Proof. Lemma 3.1 gives $v_q(\theta^2 - 2) \geq 1$. As q is odd, this forces θ to be a unit. Hensel's lemma supplies a root $\eta \in \mathbb{Q}_q$ of 2 with $v_q(\eta - \theta) = v_q(\theta^2 - 2)$, since $\eta + \theta$ is a unit. At the place $\sqrt{2} \mapsto \eta$, both summands of

$$A_i + B_i\eta = (A_i + B_i\theta) + B_i(\eta - \theta)$$

have valuation at least L_i, for every i.

Conversely choose $\theta \in \mathbb{Q}$ satisfying

$$v_q(\theta - \eta) \geq \max(\{1\} \cup \{L_i - v_q(B_i) : B_i \neq 0\}).$$

This makes the root expression have valuation at least one, and each of the two expressions tested for D_i have valuation at least L_i. The completeness assertion of Lemma 3.1, with $r = q$, now proves every required test. □

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Fix the arity k. Here are full finite definitions of the certificates. Given $\mathbf{Q} = (Q_1, \dots, Q_k)$ and $\mathbf{a} = (a_1, \dots, a_k)$, let a local witness tuple be $\mathcal{V} = (U_0, J, X_1, \dots, X_k, \theta)$. Set $Q_0 = P$, $X_0 = U_0$, and $Y_i = X_i - Q_i$ for $0 \le i \le k$. Define

$$\begin{aligned}
\mathsf{C}_K^{\mathrm{wit}}(r; \mathbf{Q}; \mathbf{a}, \mathcal{V}) \quad :\Longleftrightarrow \quad & r \neq 0 \ \wedge \ U_0, J \in G \setminus \{O\} \ \wedge \ \bigwedge_{i=1}^{k} X_i \in G \\
& \wedge \ \ell_K(z(U_0)) \neq 0 \ \wedge \ \mathsf{G}_1(r, \theta^2 - 2) \\
& \wedge \ \bigwedge_{i=0}^{k} \mathsf{L}_1(r, \theta; f(X_i)) \ \wedge \ \mathsf{L}_1(r, \theta; f(J)^\sigma) \\
& \wedge \ \bigwedge_{i=0}^{k} \mathsf{L}_1(r, \theta; f(Y_i)^\sigma) \\
& \wedge \ \bigwedge_{i=1}^{k} \mathsf{L}_K\left(r, \theta; \frac{\ell_K(z(X_i))}{\ell_K(z(U_0))} - a_i\right) \\
& \wedge \ \bigwedge_{i=0}^{k} \mathsf{L}_K\left(r, \theta; \frac{z(Y_i)^\sigma}{z(J)^\sigma}\right).
\end{aligned} \tag{16}$$

Set $\mathsf{C}_K(r; \mathbf{Q}; \mathbf{a}) \iff \exists \mathcal{V}\, \mathsf{C}_K^{\mathrm{wit}}(r; \mathbf{Q}; \mathbf{a}, \mathcal{V})$. The k free point arguments are restricted to $G \setminus \{O\}$. All entries of point tuples are rational coordinates. All tests in one occurrence share its single pair (r, θ).

Lemma B.4. *Suppose $Q_i = n_i P \neq O$ and $\mathsf{C}_K(r; \mathbf{Q}; \mathbf{a})$ holds. If $q \notin S$ and $v_q(r)$ is odd, then every a_i is q-integral and $v_q(a_i - n_i) \ge K$. If $a_i = n_i$ are nonzero integers and $r = q$ is a contact prime supplied by Hypothesis 2.4, with $q \notin S$, the certificate has rational witnesses.*

Proof. Fix witnesses. Lemma B.3 supplies one place where all listed inequalities hold. At a good place, $v(f(V)) > 0$ for $V \neq O$ means $v(x(V)) < 0$, hence V reduces to O; the zero convention also includes $V = O$. Write $U_0 = hP$, $J = jP$, and $X_i = x_i P$, with $x_0 = h$, $n_0 = 1$. The alternate quotients and (14) give

$$v_q((h-1)/j) \ge K, \qquad v_q((x_i - n_i)/j) \ge K.$$

Since j is a nonzero integer, the numerators are divisible by q^K. Thus $h \equiv 1 \pmod{q^K}$ is a unit, and

$$\frac{x_i}{h} - n_i = \frac{(x_i - n_i) - n_i(h-1)}{h} \in q^K \mathbb{Z}_q.$$

In particular x_i/h is integral. Apply (15) to X_i, U_0 and combine it with the primary test to obtain the claimed congruence and integrality of a_i. If $X_i = O$ or $Y_i = O$, the corresponding numerator is zero and the same equations hold.

For completeness, use Lemma B.2 to choose h, j at either place and put $U_0 = hP$, $J = jP$, $X_i = hn_iP$. Then $Y_i = n_i(h-1)P$ for $i \ge 1$ and $Y_0 = (h-1)P$. Multiples of formal points are formal. Lemma B.2 and (14) give $v_q((h-1)/j) \ge K$, and multiplication by each integer n_i preserves this bound. Equation (15) gives the primary inequalities with $a_i = n_i$. Finally Lemma B.3, applied to this whole finite list, provides the common rational θ and all parity witnesses. □

B.2 The height certificate

Let

$$I = \{0\} \cup (\{1,2,3\} \times (\mathcal{C} \setminus \{\infty\})), \quad c_0(\mathbf{u}, v) = v, \quad c_{(j,b)}(\mathbf{u}, v) = L_b(u_j, v).$$

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Thus $|I| = 13$. Define $\mathsf{T}^{\mathrm{wit}}_{K,k}(\mathbf{Q}; \mathbf{a}, \mathcal{W})$ as follows. Its tuple $\mathcal{W}$ consists of

$$e,\ (b_i, c_i)_{i=1}^{k},\ (u_{\nu 1}, u_{\nu 2}, u_{\nu 3}, v_\nu)_{\nu=0}^{2k},\ (r_{\nu\gamma}, \mathcal{V}_{\nu\gamma})_{0\le\nu\le 2k,\ \gamma\in I}.$$

The tuples $\mathcal{V}_{\nu\gamma}$ are fresh local witnesses for each occurrence. Put $A_0 = e$, $A_i = b_i$, $A_{k+i} = c_i$ for $1 \le i \le k$. The predicate is the following conjunction:

$$\begin{aligned} & e \neq 0, \qquad \bigwedge_{i=1}^{k} [Q_i \in G \setminus \{O\} \ \wedge\ e\,x(Q_i) = b_i + c_i\sqrt{2}], \\ & \qquad \bigwedge_{\nu=0}^{2k} [v_\nu \neq 0 \ \wedge\ u_{\nu 1} + u_{\nu 2} + u_{\nu 3} = A_\nu v_\nu], \\ & \bigwedge_{\substack{0\le\nu\le 2k \\ \gamma\in I}} \left[r_{\nu\gamma} \neq 0 \ \wedge\ \left(\bigvee_{\substack{\varepsilon\in\{\pm 1\} \\ \kappa\in\mathcal{K}}} c_\gamma(\mathbf{u}_\nu, v_\nu) = \varepsilon\kappa r_{\nu\gamma} \right) \ \wedge\ \mathsf{C}^{\mathrm{wit}}_K(r_{\nu\gamma}; \mathbf{Q}; \mathbf{a}, \mathcal{V}_{\nu\gamma}) \right]. \end{aligned} \tag{17}$$

Define $\mathsf{T}^{(k)}_K(\mathbf{Q}; \mathbf{a}) \iff \exists \mathcal{W}\, \mathsf{T}^{\mathrm{wit}}_{K,k}(\mathbf{Q}; \mathbf{a}, \mathcal{W})$. In these definitions every occurrence of Φ and every rational graph receives fresh auxiliary witnesses; these are appended to $\mathcal{W}$. Thus the definitions specify finite existential formulas, not requirements to quantify over primes. For fixed $\mathcal{W}$, its detected prime set is

$$\mathcal{P}(\mathcal{W}) = \{q \notin S :\ v_q(c_\gamma(\mathbf{u}_\nu, v_\nu)) \text{ is odd for some } \nu, \gamma\}.$$

Lemma B.5 (Normalized and raw contacts)**.** *Let $u, v \in \mathbb{Q}$, $v \neq 0$, and $(u : v) \notin \mathcal{C}$. If $q \notin S$ divides $M_S((u : v))$, then $v_q(L_b(u, v))$ is odd for some $b \in \mathcal{C}$.*

Proof. Put $t = \min(v_q(u), v_q(v))$. Every contact has valuation at least t. If $v_q(v) = t$, the infinity contact has valuation t. Otherwise $v_q(u) = t < v_q(v)$, and every finite contact has valuation t because its leading coefficient is a q-unit. Thus some contact has valuation exactly t. If t is odd, that contact suffices; if t is even, any contact of odd normalized depth has odd raw valuation. Primitive or integral coordinates are not required. □

Lemma B.6. *For every tuple $\mathcal{W}$ satisfying $\mathsf{T}^{\mathrm{wit}}_{K,k}(\mathbf{Q}; \mathbf{a}, \mathcal{W})$, write $Q_i = n_i P$ and $B = \max_i |n_i|$. If a nonzero integer D is divisible by q^K for every $q \in \mathcal{P}(\mathcal{W})$, then*

$$B^2 \le C|D|. \tag{18}$$

In particular, this applies to $D = F(\mathbf{n}) \neq 0$ if $F \in \mathbb{Z}[T_1, \dots, T_k]$ and $F(\mathbf{a}) = 0$, and to the nonzero reduced numerator of $a_i - n_i$. If $a_i = n_i \in \mathbb{Z} \setminus \{0\}$ for every i, then $\mathsf{T}^{(k)}_K(\mathbf{Q}; \mathbf{a})$ holds.

Proof. Fix $\mathcal{W}$. All its contacts are nonzero, so its slopes avoid $\mathcal{C}$. Lemma B.5 shows that every prime in any $M_S((u_{\nu j} : v_\nu))$ belongs to $\mathcal{P}(\mathcal{W})$. These radicals are squarefree, so their Kth powers divide D. Each slope has height at most $H_d|D|$. Therefore

$$h(A_\nu) \le 3H_d|D| + \log 3.$$

Since $h(\sqrt{2}) = \frac{1}{2}\log 2$, the standard sum and product inequalities, applied to $x(Q_i) = (b_i + c_i\sqrt{2})/e$, give

$$h(x(Q_i)) \le h(b_i) + h(c_i) + h(e) + \tfrac{3}{2}\log 2 \le 9H_d|D| + 3\log 3 + \tfrac{3}{2}\log 2.$$

The canonical-height lower bound and $|D| \ge 1$ give (18) with the specified C.

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For every detected prime, dividing its raw contact by $\varepsilon\kappa$ preserves valuation because κ is a unit. Lemma B.4 therefore makes all a_i integral and congruent to n_i modulo q^K. Consequently $F(\mathbf{n}) \equiv F(\mathbf{a}) = 0 \pmod{q^K}$. The same congruence implies divisibility of the reduced numerator of $a_i - n_i$, since its reduced denominator is a q-unit.

For completeness, choose a nonzero integer e clearing the rational coefficients of all $x(Q_i)$. The $2k+1$ numbers A_ν are then integers. Apply Hypothesis 2.4 independently to them, with the prime lower bound larger than every element of S. For each of the $13(2k+1)$ contacts take its positive prime as $r_{\nu\gamma}$ and use the completeness assertion of Lemma B.4 with fresh local witnesses. □

Proof of Proposition 3.2. The definitions (16) and (17), together with the point codes and the definition of G_L, give a fixed positive existential formula $\mathsf{T}_K^{(k)}$. Fix any tuple $\mathcal{W}$ witnessing this formula. Lemma B.6 gives the asserted bounds for every nonzero polynomial error and every nonzero reduced numerator. Its completeness assertion supplies witnesses when the k rational arguments are the actual nonzero integer indices. □

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